



Attention restoration during environmental exposure via alpha-theta oscillations and synchronization

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ABSTRACT

Evidence has revealed that exposure to a restorative environment can have health and cognitive benefits, but the mechanism remains unclear. This study was designed to explore the neural mechanism of environmental restorative experiences. We conducted an experiment by randomly exposing thirty-two participants to either 20 min of a restorative (wooded garden) or a nonrestorative (traffic island) environment. Participants' real-time electroencephalogram (EEG) signals were monitored via a 14-channel mobile device during the exposure. They also completed a series of psychological assessments of affective and cognitive functioning, as well as the perceived restorativeness of the environment, before and after the exposure.

The results revealed stronger and more efficient alpha-theta synchronization (functional connectivity) during the restorative experience, as well as stronger alpha-theta oscillations in the occipital lobes. Regression analysis revealed that perceived coherence was associated with the efficiency of the alpha-theta synchronization network (alpha: coef. = 2.02, 95% CI [0.68, 3.36], $p = 0.020$, $R^2 = 23.97\%$; theta: coef. = 2.41, 95% CI [1.30, 3.52], $p < 0.001$, $R^2 = 39.54\%$); fatigue recovery was associated with alpha-theta oscillations in the occipital lobes (alpha: coef. = -0.58, 95% CI [-0.88, -0.28], $p = 0.001$, $R^2 = 33.62\%$; theta: coef. = -0.50, 95% CI [-0.73, -0.26], $p < 0.001$, $R^2 = 37.74\%$); and the degree of fatigue recovery further improved attention-related cognitive performance (coef. = 0.22, 95% CI [0.03, 0.41], $p = 0.027$; $R^2 = 15.28\%$). Based on the above-mentioned evidence, we proposed that the perceived coherence of the restorative environment may induce fatigue recovery and, hence, attention restoration via alpha-theta oscillations and synchronization. The increased alpha-theta oscillations in the occipital lobes suppress visual processing, allowing the human brain to reorganize itself via alpha-theta synchronization.

1. Introduction

Exposure to nature can generate health benefits, including improving attention (Berman, Jonides, & Kaplan, 2008; Bratman, Daily, Levy, & Gross, 2015; Lee, Williams, Sargent, Williams, & Johnson, 2015; Taylor & Kuo, 2009), mitigating excessive arousal (Ulrich, 1981, 1983, 1984, 2001; Ulrich et al., 1991) and regulating emotions (McMahan & Estes, 2015). In contrast, exposure to urban environments is associated with increased mental health risks (Calderón-Garcidueñas et al., 2011; Florian et al., 2011; Marcelis, Navarro-Mateu, Murray,

Selten, & Os, 1998; Seresinhe, Preis, & Moat, 2015; Tost, Champagne, & Meyer-Lindenberg, 2015). What is the restorative mechanism of nature? Can we learn from restorative nature to design better urban environments?

Kaplan proposed that restoration results from an enjoyable and effortless cognitive appraisal of environments (R. Kaplan & Kaplan, 1989; S. Kaplan, 1995). To trigger such cognitive appraisal, an environment needs to have four characteristic components: 1) a sense of "being away" from routines or attention-consuming mental activities, 2) a sense of soft "fascination" that attracts interested viewers to linger

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without causing high levels of arousal, 3) a sense of “coherence” or “extent” when a landscape is both rich and coherent enough to allow effortless information processing, and 4) a quality of “compatibility” that affords potential desirable activities.

The four components were too vaguely defined to be applicable in design practice. Quite a few researchers have reported high inter-correlations and overlap between them (Chang, Hammitt, & Chen, 2008; Herzog, Maguire, & Nebel, 2003; Laumann, Gärling, & Stormark, 2001), even after systematic specifications in the Perceived Restorativeness Scale (PRS; Hartig, Korpela, Evans, & Gärling, 1997).

Electroencephalography (EEG) evidence on nature exposure and meditation revealed that restorative experiences may be associated with alpha and theta brain waves. In early cognitive processes, restorative natural scenery may induce more alpha oscillations (Grassini et al., 2019). These increased alpha waves, indicating possible sensory suppression (Foxe & Snyder, 2011; Toscani, Marzi, Righi, Viggiano, & Baldassi, 2010), support the hypothesis that restorative natural scenery images may occupy less attentional and cognitive resources and are, therefore, easier to process (Grassini et al., 2019; Valtchanov & Ellard, 2015). In restorative meditation, increased alpha-theta oscillations and synchronization were widely reported during practice and were correlated with anxiety reduction and mood regulation (Klimesch, 1999).

This study was designed to examine alpha-theta brain activities and their potential connections with restoration during in situ exposures. In an earlier paper (Chen, He, & Yu, 2016), we reported the differences of complex network statistics found between the two exposures, including 1/f statistics of visual stimuli and that of brain activities, small-world statistics of brain functional connectivity. In this paper, we aimed to bridge the attention restoration theory and the existing EEG evidences in environmental psychology (i.e., increasing alpha oscillations during restorative exposure) and other restorative experience (e.g., increasing alpha/theta oscillations and synchronization in music therapy and meditation). We proposed that the distinct type of information processing that occurs with exposure to a restorative environment may be related to alpha-theta activities. If so, there should be associations between alpha-theta patterns and the intuitive judgment of environmental restorativeness (i.e., PRS components), as well as the subjective feeling of fatigue recovery (Fig. 1).

2. Materials and methods

It is not easy to monitor brain activities during environmental exposure. In this study, we used mobile electroencephalography (EEG) facilitated with functional connectivity analysis (Smit, Stam, Posthuma, Boomsma, & Geus, 2008; Stam, Nolte, & Daffertshofer, 2007) to understand real-time information processing inside brains during exposure. We measured real-time neural activity during a 20-min in situ exposure using a portable wireless EEG recording device while sitting, as well as using psychological scales and a cognitive test.

2.1. Power analysis

The sample size of this study was based on the effect sizes of existing studies on restorative environments. As reported in a meta-analysis study (Bowler, Buyung-Ali, Knight, & Pullin, 2010), the impact of a restorative environment on subjectively reported fatigue revealed an effect size of 0.76 (minimal sample size = 10 calculated in G*Power 3.1

using an a priori scenario) and that for subjectively reported energy as 0.76 (minimal sample size = 10). The impact on cortisol, a biological measure of stress and nonrestorativeness, revealed an effect size of 0.57 (minimal sample size = 24). We also considered attention-related cognitive performance (effect size = 0.23), which requires a much larger sample size (196). Therefore, we decided on a sample size of 32, limiting our main objectives to subjective feelings of restoration and corresponding neural activities with only an inconclusive inference about cognitive performance. The smaller sample size allowed us to conduct a time-consuming in situ EEG experiment and functional network analysis to gain more insights into neural activity during the restorative experience.

2.2. Participants

We advertised our study to the university community via social media WeChat's Moments. Thirty-two healthy, right-handed Chinese college students volunteered to participate in this study (mean age = 20.6, SD = 1.6, 16 males). Participants were equally distributed in two conditions in terms of age, gender and time of experiment (See Supplement 1). We gave every participant a gift of ca. 30 RMB value for their participation on the next day after the experiment. This study was approved and supervised by the Ethics Committee of Tongji University (no. 2015yxy103).

Before the experiment, psychological screening, including levels of neuroticism, daily stress and recent life event stress, was performed (see Supplement 1). No significant levels of neuroticism, as measured by the NEO Personality Inventory (Costa & McCrae, 1992), were detected among participants. They reported a low level of daily stress while living in relatively stress-reducing environments. Participants were generally well rested, with a broad range of work and entertainment schedules. They were moderately exercised and were exposed to nature on a daily basis. All participants were right-handed.

2.3. Treatment: restorative/nonrestorative environments

To obtain an authentic multisensory restorative experience, we used in situ environment exposures as stimuli in this experiment. When participants fully immerse themselves in a real environment, their sensory perceptions of the environment are not compromised as they are when using picture or video stimuli.

Before the experiment, we organized a small focus group consisting of 7 landscape architecture undergrad students for a brain-storming of most restorative and nonrestorative environments on or near campus. We intentionally chose sites within a 15-min walking distance from where participants were waiting to control the impact of travelling to the sites on restorativeness.

We chose a small wooded garden by a pond as the restorative environment and a traffic island in a heavily trafficked road as the non-restorative environment (Fig. 2). There was no unpleasant direct sunlight in either site or obvious disturbance from pedestrians or insect bites. Vegetation and water were found to trigger the restorative experience, while traffic and buildings triggered the opposite (Purcell, Peron, & Berto, 2001; Tennessen & Cimprich, 1995; Ulrich, 1984). We first took a photo from the exact viewing point of where participants were sitting. We then overlaid the photos with a 20*15 grid and counted the grid cells for visibility scores. The restorative scene

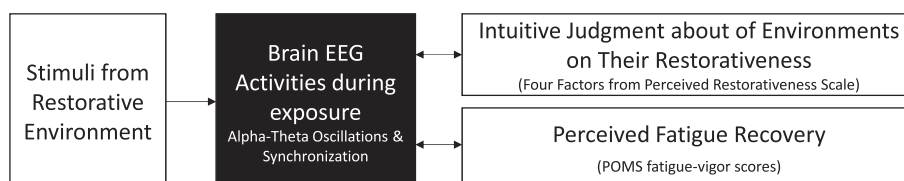


Fig. 1. A hypothetical model of alpha-theta waves and subjective measures during attention restoration.

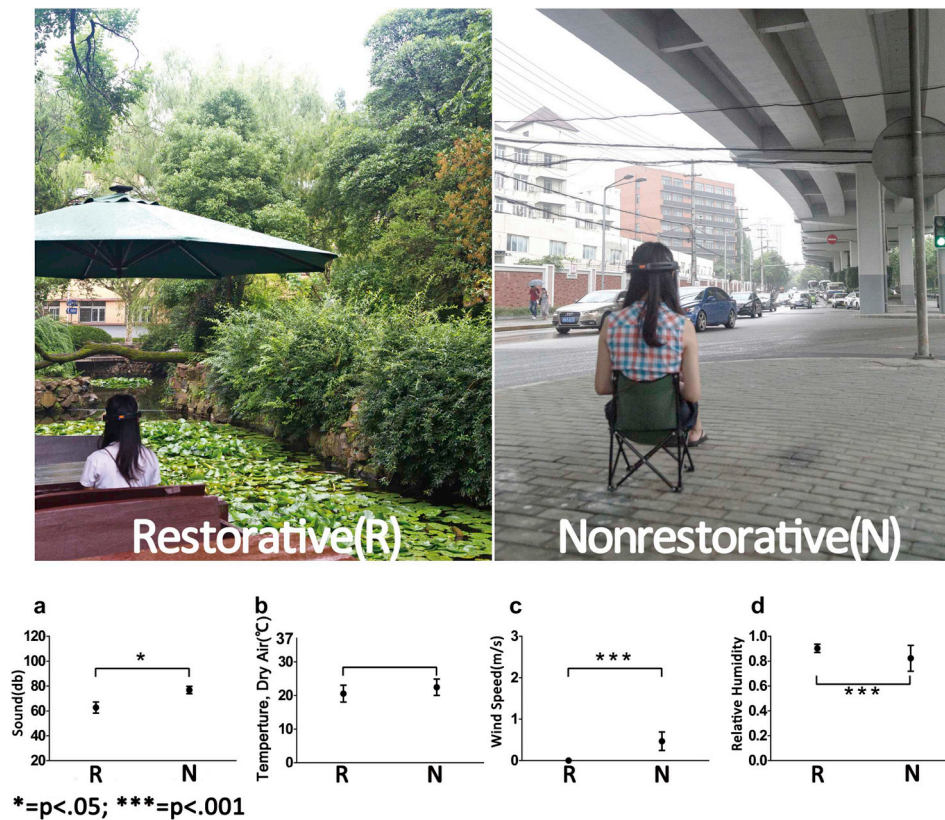


Fig. 2. The restorative (R, i.e., a wooded garden) and the nonrestorative (N, i.e., a traffic island) environments used in this study, as well as their acoustic conditions (a), temperatures (b), wind speeds (c) and relative humidities (d). Error bars depict standard error values.

consisted of 89% visible vegetation and/or water and only 4% visible buildings and paved areas, compared at 8% and 56%, respectively, in the nonrestorative scene.

To control for external variables such as thermal comfort levels, all experiments were conducted in May and June at a comfortable temperature of 22 °C (difference within ± 5), 81% humidity (difference within $\pm 13\%$), an acoustic level of 72 dB (difference within ± 12), and a wind speed of 0.8 m/s or slower (see Supplement 2).

2.4. Measures

2.4.1. EEG data collection and analysis

We used an Emotiv wireless headset (Model: Emotiv Epoc Premium, provider: EMOTIV) and its accompanying software (TestBench v1.5.1.2) to measure EEGs (Dekihara & Iwaki, 2014). The headset consisted of 14 sensors at antero-frontal (AF3, AF4, F3, F4, F7 and F8), frontocentral (FC5 and FC6), occipital (O1 and O2), parietal (P7 and P8) and temporal lobes (T7 and T8) at a sample rate of 128 Hz.

EEG signals require sophisticated analysis. Mixed weak cortical activities are recorded from the scalp and, therefore, consist of substantial amounts of noise after travelling through cerebrospinal fluid, the skull and skin. First, we discarded high/low frequency noise using filters (< 0.5 Hz or > 50 Hz) via EEGlab and then removed artifacts using Adjust 1.1 (Mognon, Jovicich, Bruzzone, & Buiatti, 2011), which is an ICA-based algorithm that automatically removes artifacts related to factors such as blood movement, temporal muscle movements, eye movements and blinks. After preprocessing, we used fast Fourier transformation to convert the signals from the time domain to the frequency domain, where frequencies were used to separate different rhythmic brain activities. Signals were categorized into five frequency bands: delta (1–4 Hz), theta (5–8 Hz), alpha (9–12 Hz), beta (13–25 Hz) and gamma (26–45 Hz). We then used functional connectivity analysis, a temporal correlation analysis between different electrodes, and a

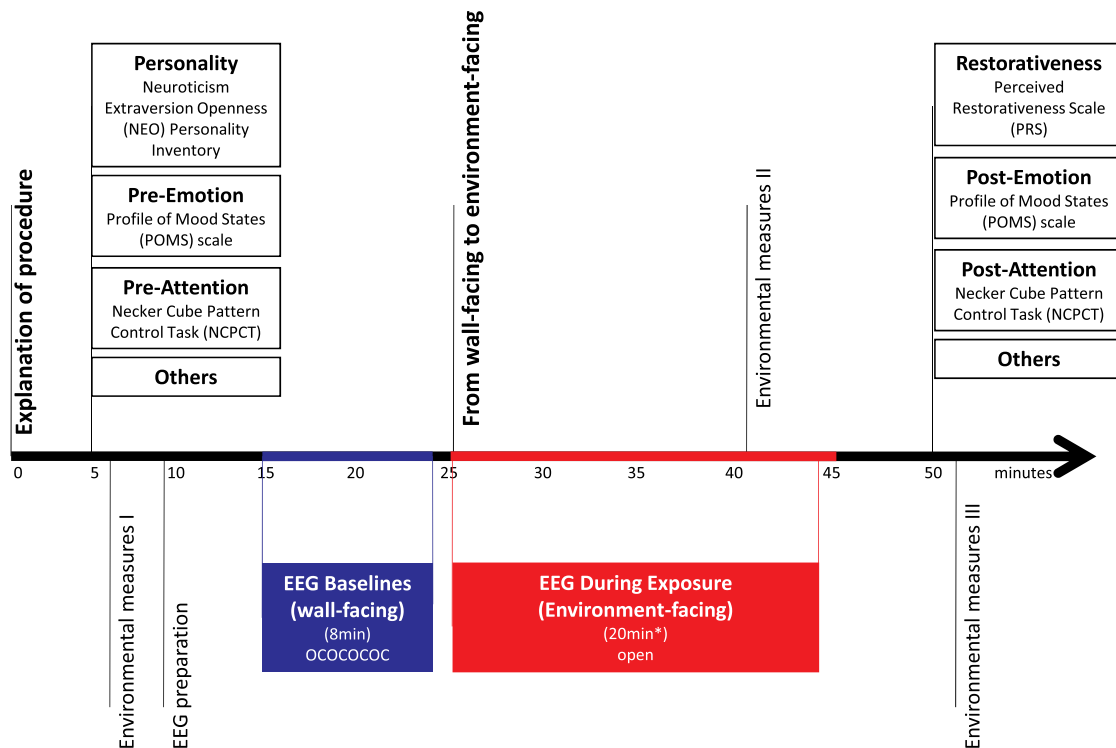
small-world network analysis to understand the brain connectivity, which is addressed in more depth in sections 2.4.2 and 2.4.3.

2.4.2. Measures of attention fatigue

2.4.2.1. Subjective feeling of fatigue. We measured the subjective feeling of fatigue using the 40-item Profile of Mood State (POMS) scale (Grove & Prapavessis, 1992), which has often been used to evaluate restorative experiences (Tsunetsugu et al., 2013; Van den Berg, Koole, & van der Wulp, 2003). Participants completed one scale upon their arrival as a pre-exposure baseline and completed another after exposure. The POMS consists of eight affective measures (tension, anger, fatigue, depression, vigor, esteem, confusion, and total mood disturbance). We adopted the changes in the two POMS scores that were relevant to restorativeness ($D_{\text{fatigue}} = \text{Post}_{\text{fatigue}} - \text{Pre}_{\text{fatigue}}$ and $D_{\text{vigor}} = \text{Post}_{\text{vigor}} - \text{Pre}_{\text{vigor}}$) and their linear combination $((D_{\text{fatigue}} - D_{\text{vigor}})/2)$ as the subjective feeling of fatigue.

2.4.2.2. Cognitive performance of attention fatigue. We measured the cognitive performance of attention fatigue using the Necker Cube Pattern Control Test (NCPCT), a cognitive test that measures the frequency one fails to maintain concentration (Orbach, Ehrlich, & Heath, 1963). The NCPCT has been shown to be a reliable measure for attention fatigue in restorativeness studies (Hartig, Evans, Jamner, Davis, & Gärling, 2003; Tennessen & Cimprich, 1995), with a smaller score indicating a higher attention level. A baseline measure was collected before environmental exposure, and a postexposure measure was collected afterwards. The difference in the NCPCT scores ($D_{\text{NCPCT}} = \text{Post}_{\text{NCPCT}} - \text{Pre}_{\text{NCPCT}}$) was used to measure the impact of the environmental exposure on attention-related cognitive performance.

2.4.2.3. EEG measures of relaxed attention. We used the power in alpha and theta EEG oscillations to measure relaxed attention during exposure. An earlier study found higher alpha power during a more



* Initially, we only recorded the EEG during the latter 10 minute exposure for the first 16 participants (8 restorative group). We later extended our recording to the entire 20-minute for the latter 16 participants (8 restorative group). Paired t-tests revealed no significant difference between the first 10 minutes and the latter 10 minutes in terms of EEG correlation (based on which we calculated functional connectivity network statistics) and β slope (the power ratio of high frequencies and low frequencies) for the 16 participants with entire EEG record except for two participants in nonrestorative group. Therefore, the full recording length from both groups (10 minutes for the first 16 and 20 minutes for the latter 16) were used for analyses.

Fig. 3. Experimental procedure.

relaxing and restorative experience (Ulrich, 1981). A recent event-related study confirmed that an increase in alpha and theta power, especially at the occipital lobes, was associated with reduced attention-related arousal (Lin et al., 2010). In this study, we analyzed the alpha and theta power of all electrodes, especially the two (O1, O2) electrodes at the occipital lobes. We used the averaged alpha and theta power of the O1 and O2 electrodes as indicators of relaxed attention during exposure.

2.4.3. Measures of restorativeness

2.4.3.1. Subjective restorative factors. The restorativeness of the environmental experience was measured via the Perceived Restorative Scale (PRS) developed by Hartig et al. (1997) based on Kaplan's attention restoration theory (R. Kaplan & Kaplan, 1989; S. Kaplan, 1995). The original PRS scale consists of 16 items that measures four mental constructs: being away, fascination, coherence and compatibility. A focus group discussion before the experiment found that the direct translations were confusing. Therefore, we added explanations to two coherence items and removed three compatibility items (see Supplement 3). All items were measured on a seven-degree Likert scale ranging from -3 (very much disagree) to 3 (very much agree).

2.4.3.2. EEG measures of restorativeness. Brain functional connectivity refers to the synchronization between electrodes above different brain regions, recording "how neurons communicate" and revealing important aspects of attention and cognitive processes (Salinas & Sejnowski, 2001). Improved human affective regulation and cognitive performance related to restorative experiences have been found to be associated with enhanced functional connectivity, especially EEG synchronization in the theta and alpha bands (Cahn & Polich, 2006; Travis & Wallace, 1999) and its efficiency (Xue, Tang, Tang, & Posner, 2014).

The restorative effect during restorative meditation was found to be associated with higher small-world network characteristics in brain functional connectivity (Xue et al., 2014), especially in the lower frequencies such as the theta and alpha bands (Klimesch, 1999; Marzetti et al., 2014; Xue et al., 2014). Small-world network characteristics describe a type of efficient communicating network that is widely found in biological, ecological, social, internet, molecular and neuronal networks (Watts & Strogatz, 1998). A well-connected, efficient brain network often simultaneously has both a higher functional segregation (larger clustering coefficient, i.e., C_{mean}) and a higher functional integration (smaller average shortest path length, i.e., L_{mean}) (Rubinov & Sporns, 2010); the former enables sufficient task specialization, while the latter facilitates adequate coordination between these divisions, leading to efficient information processing and, therefore, increasing restorativeness.

We used a functional efficiency index in the alpha and theta bands to describe restorativeness. To compute the EEG functional network, a weighted matrix is usually calculated first by the signal coherence between electrodes and is then reduced to a sparse binary network via multilevel thresholding (Bullmore & Sporns, 2009; Rubinov & Sporns, 2010). The selection of a threshold should preserve the strongest connections of the weighted network, although this process is often arbitrary (Rubinov & Sporns, 2010; Tomasi & Volkow, 2011). We examined the network connectivity at correlation coefficient thresholds ranging from 0.05 to 0.95 at a step of 0.05. We adopted a range of thresholds to minimize the arbitrariness of a single one. Specifically, we used all the mid-range (from 0.55 to 0.75) where the efficient connection were evident. Then, we calculated the connectivity small-world-network efficiency index (C_{mean}/L_{mean}) of each participant. The connectivity index is a combination of the average shortest path length (L_{mean}) and clustering coefficient (C_{mean}).

2.5. Procedure

The entire experimental procedure took approximately 50 min (Fig. 3). The 32 participants were first randomly assigned to either the restorative exposure group ($n = 16$) or the nonrestorative exposure group ($n = 16$). Upon their arrival, participants signed a written consent form after reading the explanation of the experimental procedure and risks. They then completed a two-page pretest questionnaire followed by an NCPCT test. After placing a 14-electrode portable EEG headset on their heads, participants were asked to sit facing a wall, which temporarily blocked the visual information they were processing, while we took their pre-exposure EEG baseline measurements. They were instructed to alternate between either opening or closing their eyes in four 1-min cycles during the pretest EEG baseline acquisition.

After pretest baseline measurements were obtained, participants were instructed to turn and face the environment, either a restorative garden or a nonrestorative traffic island, for 20 min. To prevent mind-wandering and drowsiness, participants were instructed to count from 1 to 1000 soundlessly at a speed comfortable for them without interrupting the environmental experience. After environmental exposure, participants completed a postexposure questionnaire, a short interview about their experience and the NCPCT test again.

2.6. Statistical analysis

We compared the differences between restorative and non-restorative environments using independent *t*-tests in this between-subjects design. We tested the means of each condition away from a neutral zero point, i.e., the middle point of a Likert scale or true zero changes, using one-sample *t*-tests.

We examined the associations between EEG and subjective measures using linear regression. Although only two conditions were used, participants responded to the stimuli at different intensity levels. If they were valid bioindicators for the neural mechanism of restorative experience, the EEG measures should be quantitatively associated with the degree of perceived restorativeness or fatigue both within and between groups when taking individual sensitivity into consideration. Therefore, we adopted linear regression in addition to mean difference tests to exploit both the between-group and within-group variances.

T-tests and descriptive statistics were performed in SPSS 21, while regression and correlation analysis were performed in Stata 15.0. A significance level of 0.05 was used in all statistical tests.

3. Results

3.1. Descriptive statistics

We have reported the descriptive statistics and normality test results of all interested variables under the two conditions in Table 1.

3.2. Attention fatigue

3.2.1. Subjective feelings of fatigue and cognitive performance

Self-reported vigor ($D_{\text{vigor}} = \text{Post}_{\text{vigor}} - \text{Pre}_{\text{vigor}}$) was found to be more decreased after nonrestorative environment exposure than after restorative exposure ($\Delta = 3.58$, 95% *CI* [0.08, 7.07]; $t(30) = 2.09$, $p = 0.045$), as shown in Fig. 4. In contrast, self-reported fatigue ($D_{\text{fatigue}} = \text{Post}_{\text{fatigue}} - \text{Pre}_{\text{fatigue}}$) was found to increase more ($\Delta = -4.06$, 95% *CI* [-7.45, 0.68]; $t(30) = -2.45$, $p = 0.020$). The linear combined fatigue-vigor measure ($D_{\text{fatigue-vigor}} = D_{\text{fatigue}} - D_{\text{vigor}}$) revealed consistent between-group differences ($\Delta = -3.82$, 95% *CI* [-6.75, -0.89]; $t(30) = -2.66$, $p = 0.012$). A one-sample test revealed significantly increased fatigue after nonrestorative exposure ($\Delta = 2.88$, 95% *CI* [0.75, 5.01]; $t(15) = 2.88$, $p = 0.011$) but no significant changes after non-restorative exposure ($\Delta = -3.94$, 95% *CI* [-3.14, 1.26]; $t(15) = -0.91$, $p = 0.378$).

Table 1

Descriptive statistics of interested variables.

	Non-restorative		Restorative	
	Mean	SD	Mean	SD
Attention Fatigue				
Subjective feeling of fatigue				
D_{Fatigue}	-2.38	4.65	1.69	4.73
D_{Vigor}	-0.50	4.65	-4.08	5.02
$D_{\text{fatigue}} - D_{\text{Vigor}}$	-1.88	8.25	5.76	7.99
Cognitive performance				
D_{NCPCT}	-0.75	3.15	0.31	2.09
EEG measures of relaxed attention				
O12 alpha	3.81	5.15	1.05	3.18
O12 theta	6.09	6.70	2.76	3.42
Restorativeness				
Subjective restorativeness				
Fascination	0.05	0.48	-0.56	0.59
Coherence	1.25	1.12	0.08	1.08
BeingAway	0.57	0.75	-0.59	0.76
Compatibility	0.97	1.43	-2.03	1.09
EEG restorativeness				
connectivity theta	0.78	0.02	0.72	0.04
connectivity alpha	0.79	0.03	0.73	0.04
$C_{\text{mean}}/L_{\text{mean}}$ theta	0.72	0.30	0.45	0.33
$C_{\text{mean}}/L_{\text{mean}}$ alpha	0.75	0.26	0.55	0.32

The cognitive performance of attention fatigue, which was measured by changes in NCPCT test scores ($D_{\text{NCPCT}} = \text{Post}_{\text{NCPCT}} - \text{Pre}_{\text{NCPCT}}$, an increase indicating reduced cognitive performance and increased attention fatigue), revealed no significant difference between restorative and nonrestorative exposures ($\Delta = -0.62$, 95% *CI* [-2.49, 1.11]; $t(27.6) = -0.688$, $p = 0.440$), despite the potential trend implied in Fig. 4.

3.2.2. EEG measures of relaxed attention

During the restorative experience, a higher posterior theta-alpha power was observed, especially in the occipital lobes. We analyzed the mean power in alpha and theta bands during the restorative and the nonrestorative experience at the 14 electrodes and mapped the mean power on the brain. As shown in Fig. 5, EEG theta-alpha power was higher in the medial-to-posterior side of the brain during the restorative experience than during the nonrestorative experience, especially in the occipital lobe (O1 and O2). An independent *t*-test revealed a difference, though not significant, in the occipital lobes in the alpha ($\Delta = 2.75$, 95% *CI* [-0.33, 5.84]; $t(30) = -1.82$, $p = 0.079$) and theta ($\Delta = 3.34$, 95% *CI* [-0.56, 7.23]; $t(22.3)$, equal variances not assumed) = -1.77, $p = 0.090$) bands.

3.2.3. Associations between subjective fatigue, cognitive performance and EEG

The results revealed a strong negative association between alpha-theta oscillations in the occipital lobe and the subjective feeling of fatigue, while the association between this latter measure and attention-related cognitive performance was inconclusive (Fig. 6).

Regression analyses found that increased theta oscillations in the occipital lobe (measured by EEG electrodes at O1 and O2) can predict a fair amount of the combined fatigue-vigor ($D_{\text{fatigue-vigor}}$) measure (coef. = -0.50, 95% *CI* [-0.73, -0.26], $p < 0.001$; $R^2 = 37.74\%$, adjust $R^2 = 35.67\%$, Fig. 6-a), as well as the separate fatigue (coef. = -0.49, 95% *CI* [-0.78, -0.20], $p = 0.002$; $R^2 = 28.58\%$, adjust $R^2 = 26.20\%$) and vigor (coef. = 0.50, 95% *CI* [0.21, 0.79], $p = 0.001$; $R^2 = 29.28\%$, adjust $R^2 = 26.92\%$) measures. Similar associations were found between the combined fatigue-vigor ($D_{\text{fatigue-vigor}}$) measure and the alpha oscillations in the occipital lobe (coef. = -0.581, 95% *CI* [-0.89, -0.28], $p = 0.001$; $R^2 = 33.62\%$, adjust $R^2 = 31.41\%$; Fig. 6-b) and in the two separate measures.

The combined fatigue-vigor ($D_{\text{fatigue-vigor}}$) measure, as regression revealed, was further related to attention-related cognitive performance

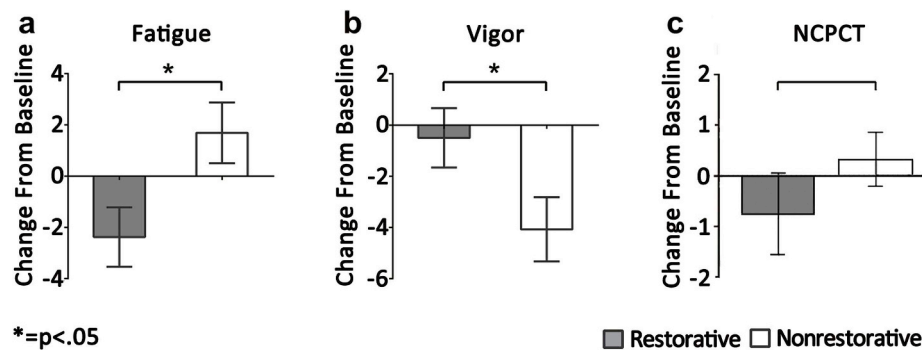


Fig. 4. The changes in subjective feelings of fatigue measured by the PMOS fatigue score (a) and vigor score (b), as well as the changes in cognitive performance as measured by the NCPCT attention test (c) after 20-min restorative and nonrestorative exposures. Error bars depict standard error values.

(D_{NCPCT} ; coef. = 0.22, 95% CI [0.03, 0.41], $p = 0.027$; $R^2 = 15.28\%$, adjust $R^2 = 12.46\%$; Fig. 6-c).

3.3. Restorativeness

3.3.1. Subjective restorativeness

All four subjective restorativeness factors of the attention restoration theory (R. Kaplan & Kaplan, 1989; S. Kaplan, 1995), as measured by the PRS (Hartig et al., 1997), i.e., a sense of being away ($\Delta = 1.16$, 95% CI [0.61, 1.70]; $t(30) = 4.31$, $p < 0.001$), fascination ($\Delta = 0.61$, 95% CI [0.22, 1.00]; $t(28.9)$, equal variances not assumed) = 3.23, $p = 0.003$), coherence ($\Delta = 1.17$, 95% CI [0.37, 1.96], $t(30) = 3.00$, $p = 0.005$) and compatibility ($\Delta = 3.00$, 95% CI [2.08, 3.92], $t(30) = 6.68$, $p = 0.005$), were perceived to be higher in the restorative experience than in the nonrestorative experience (Fig. 7).

A one-sample test comparing t restorative/nonrestorative against the neutral condition (scores = 0) specified that the restorative

environment in this study was perceived as providing a sense of coherence ($\Delta = 1.25$, 95% CI [0.65, 1.85]; $t(15) = 4.47$, $p < 0.001$), an enjoyable escape (i.e., being away, $\Delta = 0.57$, 95% CI [0.17, 0.97]; $t(15) = 3.03$, $p = 0.009$) and compatibility ($\Delta = 0.97$, 95% CI [0.21, 1.73]; $t(15) = 2.71$, $p = 0.016$), but was not necessarily fascinating ($\Delta = 0.05$, 95% CI [-0.21, 0.31]; $t(15) = 0.415$, $p = 0.684$); the nonrestorative environment, i.e., a traffic island, was perceived as being incompatible ($\Delta = -2.03$, 95% CI [-2.61, -1.45], $t(15) = -7.47$, $p = 0.001$), boring (i.e., having low fascination, $\Delta = -0.56$, 95% CI [-0.87, -0.25]; $t(15) = -3.85$, $p = 0.002$), and was, apparently, the opposite of an enjoyable escape (i.e., being away, $\Delta = -0.59$, 95% CI [-0.99, -0.18]; $t(15) = -3.07$, $p = 0.008$), but was not necessarily incoherent ($\Delta = 0.08$, 95% CI [-0.49, 0.66]; $t(15) = 0.309$, $p = 0.761$).

3.3.2. EEG restorativeness: alpha-theta connectivity efficiency indices

We illustrated brain activity in functional connectivity maps in alpha-theta bands. As we can see from those maps, stronger connections

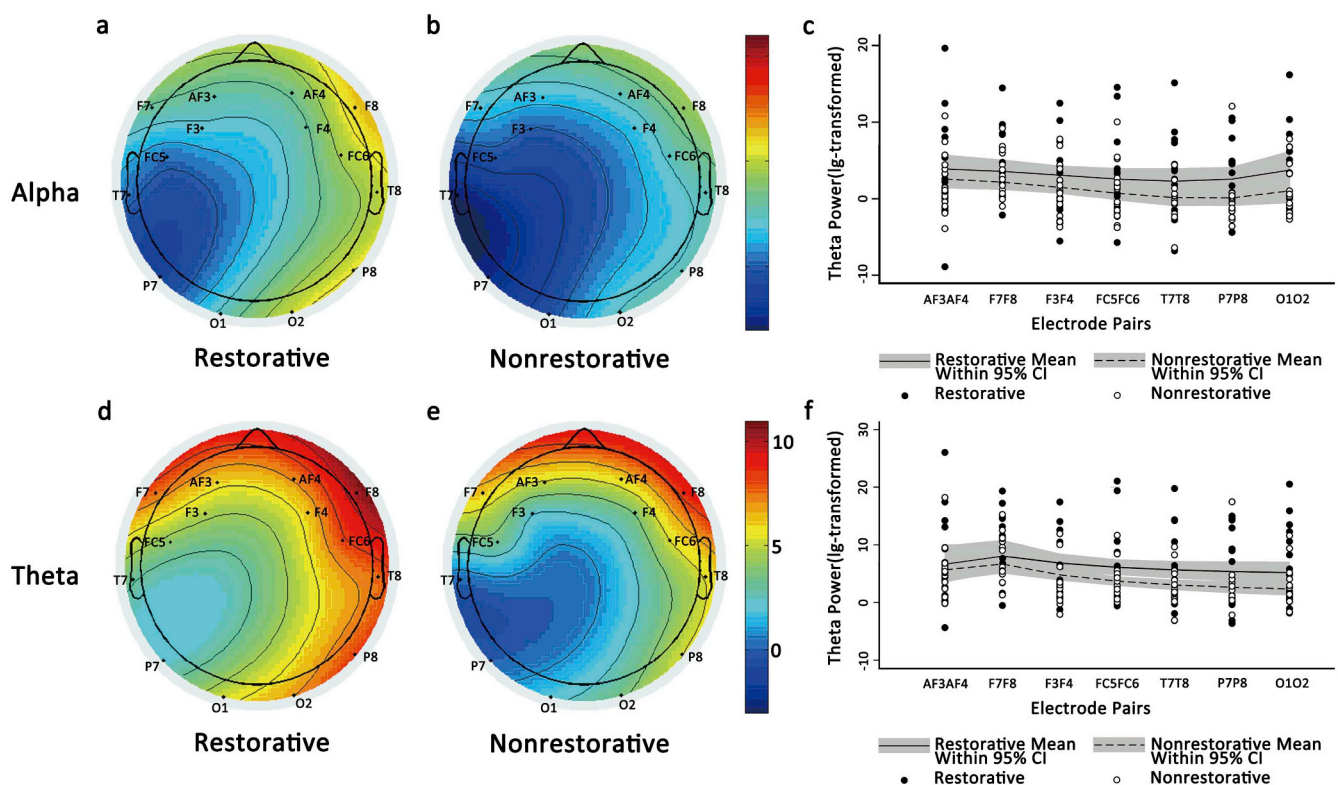


Fig. 5. The alpha and theta oscillations across the brain during restorative and nonrestorative experiences. Alpha power maps during restoration and the nonrestorative experiences were illustrated in a and b, respectively. A scatter plot (c) revealed the from-front-to-back alpha power differences across participants in the two experience conditions, with a fractional-polynomial prediction fit. Similar power maps and scatter plot in the theta band are shown in d, e and f.

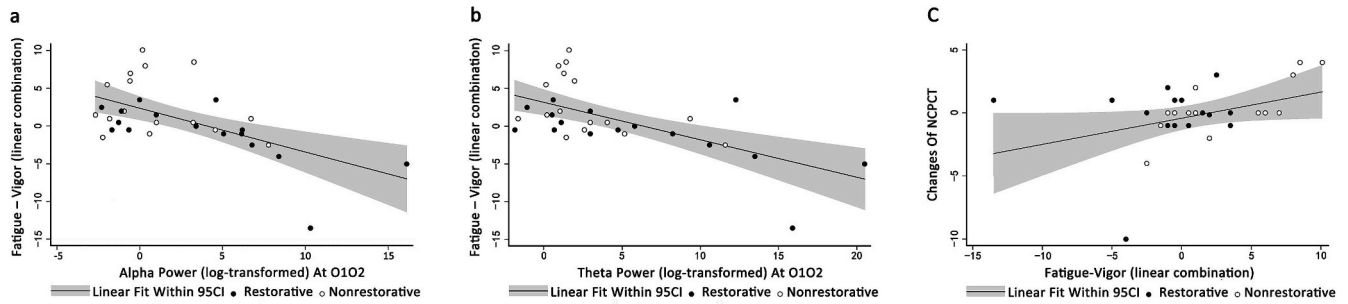


Fig. 6. Associations between subjective fatigue, cognitive performance and EEG measures. The variance in the linearly combined POMS fatigue-vigor measure ($(D_{\text{fatigue}} - D_{\text{vigor}})$) of the restorative experience group could be fairly explained by the alpha (a, $\text{coef.} = -1.16$, 95% $CI [-1.77, 0.55]$, $p < 0.001$; $R^2 = 33.62\%$) and theta (b, $\text{coef.} = -0.80$, 95% $CI [-1.34, -0.26]$, $p = 0.006$; $R^2 = 42.26\%$) oscillations (measured by the averaged log-transfer power of at O1 and O2 EEG electrodes). No significant association between the same variables was observed in the nonrestorative group (alpha: $\text{coef.} = -0.69$, 95% $CI [-2.07, 0.69]$, $p = 0.301$; theta: $\text{coef.} = -1.06$, 95% $CI [-2.25, 0.12]$, $p = 0.075$). The combined fatigue-vigor ($D_{\text{fatigue-vigor}}$) measure might be further related to attention-related cognitive performance (c, $\text{coef.} = 0.10$, 95% $CI [0.00, 0.21]$, $p = 0.055$).

were seen in the restorative environment exposure (Fig. 8-a, d) than in the nonrestorative scenario (Fig. 8-b, e), both within the frontal lobes and from the frontal lobes to other regions. The right frontotemporal region, in particular, seemed to function as a hub during the restorative experience. These seemingly stronger connections in the maps may indicate more efficient brain connectivity during exposure to a restorative environment.

We tested the connectivity efficiency theory by comparing the averaged mid-range connectivity indices ($C_{\text{mean}}/L_{\text{mean}}$) from correlation coefficient thresholds of 0.55–0.75. The restorative experience induced significantly more efficiency in the theta band ($\Delta = 0.25$, 95% $CI [0.37, 0.46]$; $t(30) = 2.40$, $p = 0.023$, Fig. 8-f) than that induced by the nonrestorative experience, but this effect was not significant in the alpha band ($\Delta = 0.54$, 95% $CI [-0.23, 0.36]$; $t(30) = 1.80$, $p = 0.082$, Fig. 8-c).

3.3.3. Associations between subjective restorativeness and alpha-theta connectivity efficiency indices

The results revealed a strong positive association between the perceived coherence and brain connectivity indices (Fig. 9). A linear regression analysis found that the general brain connectivity index in the theta band could predict 39.54% ($\text{adj } R^2 = 37.53\%$) of the variance ($\text{coef.} = 2.41$, 95% $CI [1.30, 3.52]$, $p < 0.001$, Fig. 9-a) in the perceived coherence as measured by the PRS coherence score. The connectivity in the alpha band could predict 23.97% ($\text{adj } R^2 = 21.44\%$) of the same perceived coherence ($\text{coef.} = 2.02$, 95% $CI [0.68, 3.36]$, $p = 0.020$, Fig. 9-b). No association was found between the connectivity index and the other three PRS scores (i.e., a sense of being away, fascination or compatibility).

4. Discussion

The results indicate that alpha-theta activity may play an important role in restoration, which advances our understanding of Kaplan's attention restoration theory (Fig. 10). During exposure to a restorative environment, people may perceive a sense of coherence within the environment, which results in increased alpha-theta synchronization in an efficiently communicating network within the brain. Alpha-theta synchronization is a top-down restorative process that occurs throughout the brain and that is accompanied by increased alpha-theta oscillations in the occipital lobes, which suppresses lower visual inputs. The reduced load in external visual information processing allows the brain to focus more on self-restoration, which eventually leads to fatigue recovery and, consequently, improved attention-related cognitive performance.

4.1. Fatigue recovery and alpha-theta oscillations

Exposure to restorative landscapes helps people recover from fatigue and restore their attention (R. Kaplan & Kaplan, 1989; S. Kaplan, 1995, 2001). Although the postexposure effect of attention fatigue recovery is supported by substantial empirical evidence (Bowler et al., 2010; Ohly et al., 2016; Velarde, Fry, & Tveit, 2007), very little is known about what happens during environmental exposure except a few in situ experiments with real-time biomeasures (Chang & Chen, 2005; Park et al., 2007; Tsunetsugu et al., 2013; Ulrich, 1981). This study extended our knowledge about real-time in situ brain activities.

We observed higher power alpha-theta oscillations in the occipital lobes during a restorative experience, which may indicate less awareness of the environment. The lower visual cortex is located in the occipital lobes and uses visual information from sensory receptors to perform primary processing before sending this information to areas of

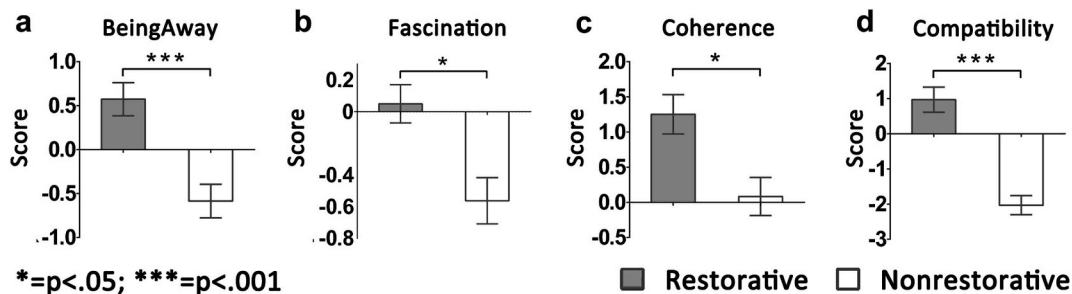


Fig. 7. Self-reported nature experience, as measured by the Perceived Restorativeness Scale (PRS). We reduced the 13-item scores to the four mental constructs in the PRS – being away (a), fascination (b), coherence (c), and compatibility (d) – by averaging. All constructs ranged from –3 (very much disagree) to 3 (very much agree). Error bars depict standard error values.

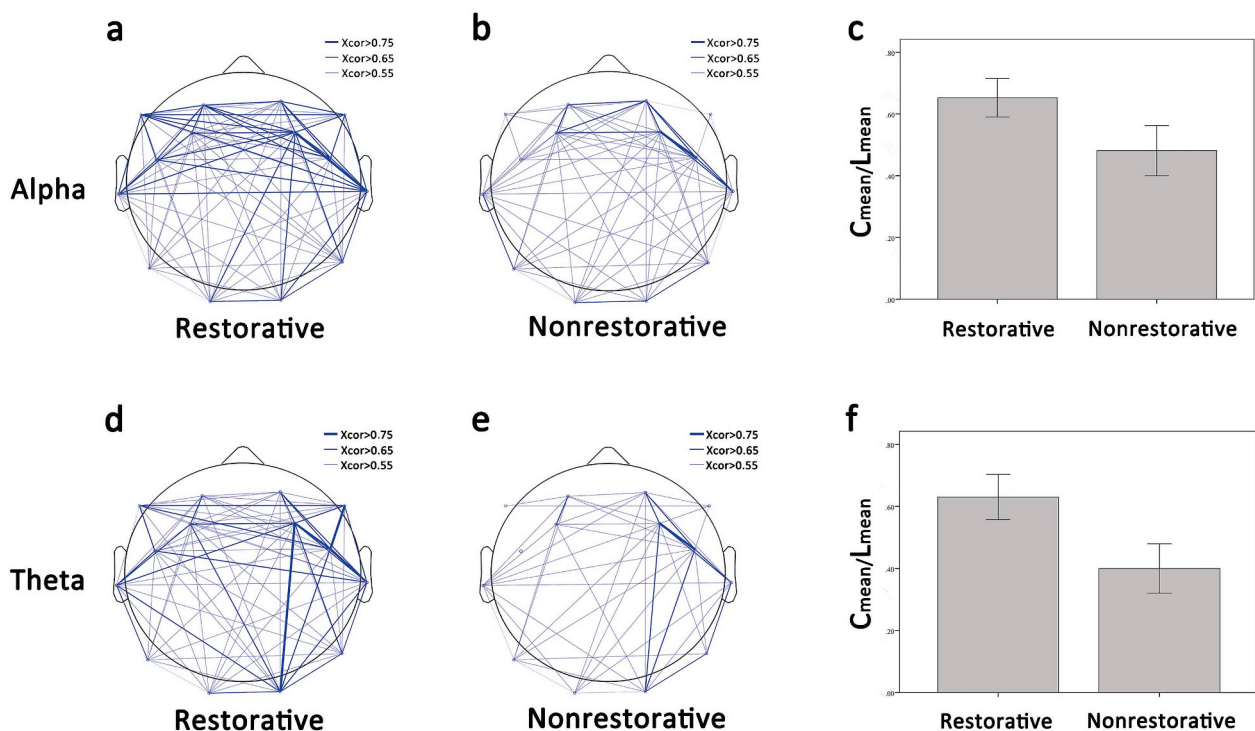


Fig. 8. EEG functional connectivity network statistics. The functional connectivity networks during the restorative and nonrestorative exposures in the alpha band were illustrated in *a* and *b*, respectively, while their corresponding efficiency indices (C_{mean}/L_{mean}) were illustrated in *c*. The same networks and indices in the theta bands were presented in the same order below in *d*, *e* and *f*. Error bars in *c* and *f* depict standard error values.

higher cognition (Kandel, Schwartz, Jessell, Siegelbaum, & Hudspeth, 2013). An increase in the power of alpha-theta oscillations in the occipital lobes may be associated with lower alertness when responding to environmental visual signals (Lin et al., 2010). In other words, our EEG evidence indicates that the participants exposed to the nonrestorative traffic environment were more vigilant, preparing to respond to impromptu environmental information.

Our observations of increased EEG alpha oscillations associated with fatigue recovery during the exposure to restorative environment were consistent with the observations in the existing literature (Chang & Chen, 2005; Ulrich, 1981). We found that restorative exposure not only induces stronger alpha oscillations but also stronger theta oscillations, especially in the occipital lobes. Similarly increased alpha-theta oscillations have been observed during meditation (Cahn & Polich,

2006; Hagerty et al., 2013), which is believed to trigger similar restorativeness (S. Kaplan, 2001).

During restorative environment exposure, increased alpha-theta power in the occipital lobes may suppress alertness and, therefore, might save cognitive resources for fatigue recovery. Our results revealed a strong positive association between alpha-theta oscillations in the occipital lobes and perceived fatigue recovery.

4.2. Perceived coherence and alpha-theta synchronization

Perceived environmental coherence is an important factor in attention restoration (Hartig et al., 1997; R.; Kaplan & Kaplan, 1989). Environmental coherence reflects orderly organization with simple distinct regions, repeating themes and unifying textures (R. Kaplan,

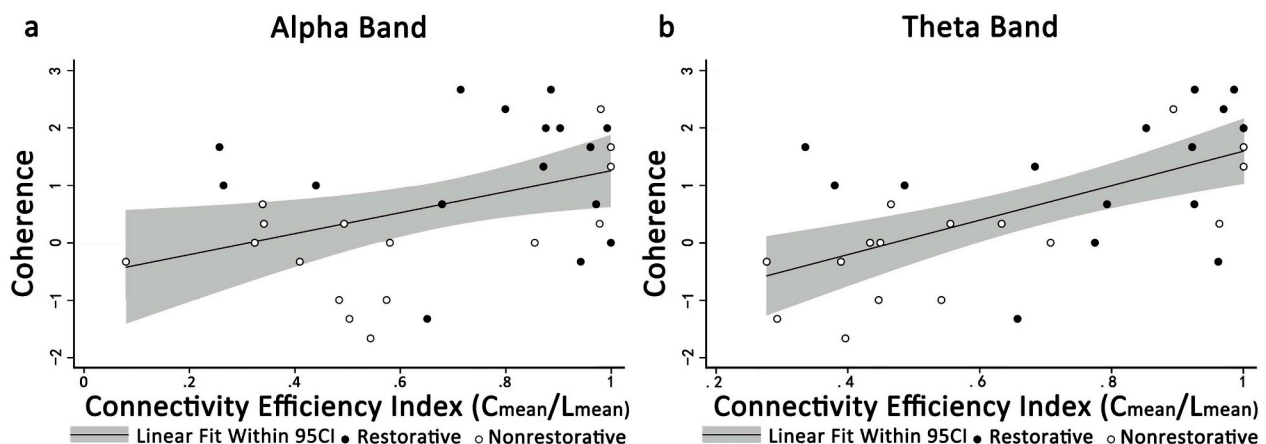


Fig. 9. The variance in the perceived environmental coherence (measured by the PRS coherence score) of the nonrestorative group can be partially explained by the brain functional connectivity efficiency index (C_{mean}/L_{mean}) in the alpha (*a*, coef. = 2.02, 95% CI [0.68, 3.36], $p = 0.020$, $R^2 = 23.97\%$) and theta bands (*b*, coef. = 2.41, 95% CI [1.30, 3.52], $p < 0.001$, $R^2 = 39.54\%$). No significant association between the same variables was observed in the restorative group (alpha: coef. = 0.43, 95% CI [-2.12, 2.97], $p = 0.726$; theta: coef. = 1.31, 95% CI [-0.76, 3.38], $p = 0.195$).

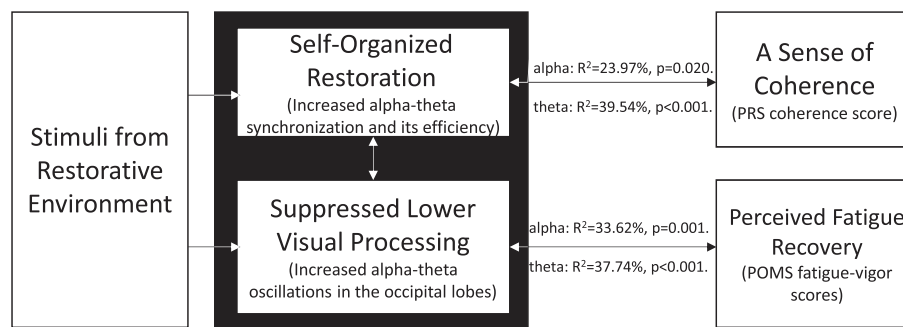


Fig. 10. A revised hypothetical model of alpha-theta waves and subjective measures during attention restoration.

Kaplan, & Ryan, 1998). However, we know very little about how coherent landscapes induce restorative experiences such as recovery from fatigue or a reduction in negative emotions. In this study, we found that perceived environmental coherence was positively associated with the efficiency of alpha-theta synchronization, indicating efficient neuron communication. We identified an association between real-time brain activities and coherence, which confirmed and advanced existing findings suggesting that spatial statistics can influence human cognition and preferences (Kardan et al., 2015).

We observed more efficient alpha-theta synchronization during the restorative experience than during the nonrestorative experience. People are more likely to find an environment to be more chaotic when their brains communicate less efficiently. Decreased efficiency in the alpha-theta synchronization was found to be associated with reduced perceived coherence in the environment.

We observed enhanced and more efficient alpha-theta synchronization across the whole brain during restorative environmental exposure. Increased alpha-theta synchronization improves connections across the whole brain. Visible connecting clusters, such as those in the frontal lobes, especially in the right dorsolateral prefrontal area (F4), were strengthened during restorative environmental exposure. The dorsolateral prefrontal area plays an important role in attention restoration (S. Kaplan & Berman, 2010). Neurological evidence has revealed that theta synchronization between the prefrontal cortex, amygdala and hippocampus is important in negative emotion regulation (Ertl, Hildebrandt, Ourina, Leicht, & Mulert, 2013). Increased alpha-theta synchronizations in the frontal lobes may help regulate emotions. Reduced negative emotions and enhanced positive emotions were often observed during restorative experiences (Bodin & Hartig, 2003; Bowler et al., 2010; Bratman et al., 2015).

The increased alpha-theta synchronization and efficiency observed after a restorative experience in this study may indicate self-organized restoration in human brains, which is consistent with effortless attention recovery in Kaplan's (1995) theory. During a restorative experience, the brain may restore and optimize itself via an alpha-theta synchronization network. Alpha and theta oscillations usually reflect cognitive and memory performance (Klimesch, 1999). Synchronization in the alpha rhythm may reflect top-down, inhibitory control processes (Klimesch, Sauseng, & Hanslmayr, 2007), while that in the theta rhythm could reveal new information encoding (Klimesch, 1999). During a restorative experience, the human brain may restore and optimize itself via an alpha-theta synchronization network. Similarly increased small-world statistics in the alpha-theta synchronization were observed during mindful meditation (Xue et al., 2014).

4.3. Potential implications for design and planning

The findings provide supporting evidences that the coherent designs of our environments and buildings are not only pleasant, but also can be beneficial to our health. Nature is full of complex details. Despite the complexity, many natural landscapes still appear harmonious and

coherent to us. It seems that our brains are smart in sorting the complex information from nature environment. Somehow the information process of restorative nature is more efficient, and therefore does not over occupy our brains. Our brains seem to evaluate the "coherence" of an environment to decide whether we should be more alert of our immediate environment or not. We begin to realize that coherence, which was commonly viewed as an aesthetic concern only, may influence information processing and, therefore, bring health benefit like restoration. Incoherent environments, in contrast, can require more cognitive resources to process, and therefore, prolonged exposure to them may cause fatigue.

4.4. Limitations and future research directions

We cannot specify which environmental stimuli any one particular participant was paying attention to during the environmental exposure because in situ environmental exposure was adopted to capture the holistic multisensory experience. When exposed to the same non-restorative environment, one participant may feel nothing watching cars come and go, while another participant might get quite annoyed. We only examined two sites, but the individual difference above may aggravate the variance (Figs. 5 and 8). It is important to note that the statistical association does not imply causation, and verifications are needed to examine more sites in the future.

It would be valuable in future research to examine how we can improve the restorativeness of a constructed environment without compromising its function. Much evidence has confirmed that vegetated environments are more restorative than barren urban ones (Bowler et al., 2010; Capaldi, Dopko, & Zelenski, 2014; Maller, Townsend, Pryor, Brown, & St Leger, 2006; Ohly et al., 2016; Thompson Coon et al., 2011; Velarde et al., 2007), but very few have explored the variance within built environments (e.g., Herzog et al., 2003). With real-time biomeasures such as EEG, the nuanced differences within built environments could be better examined.

5. Conclusions

Our study contrasted the impacts of two types of environmental exposures — one restorative (a wooded garden) and one nonrestorative (a traffic island) — on human restorative experiences and cognitive processes via an in situ experiment using a combination of psychological and neurological measures. The results revealed increased alpha-theta brain oscillations in the occipital lobes during a restorative experience, as well as better connected alpha-theta synchronization across the whole brain, which were associated with fatigue reduction and a sense of coherence, respectively.

CRediT authorship contribution statement

Zheng Chen: Data curation, Formal analysis, Writing - original draft, Writing - review & editing. **Yujia He:** Data curation, Formal

analysis. **Yuguo Yu:** Writing - review & editing.

Declaration of competing interest

We declare no conflicting interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvp.2020.101406>.

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